BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, DG KHAN

INTERMEDIATE (11TH CLASS) EXAMINATION, 2024

MATHEMATICS

Past Paper 2024 • Paper Code: 6197 / 6198

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

OBJECTIVE

Maximum Marks: 20 | **Paper Code:** 6197

Note: You have four choices for each objective type question as A , B , C and D . The choice which you think is correct, fill that circle in front of that question number. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question.

Q1. Factorial form of (n+1) (n) (n-1) is

(A) (n+1)!/(n-2)!
(B) (n-2)!/(n+1)!
(C) (n+1)!/n!
(D) n!/(n+1)!

Q2. $(2 + i)^2 \div (2 - i)^2$ = ----------

(A) 4 i
(B) 8 i
(C) 6 i
(D) 10 i

Q3. Value of $\sin^2 \pi / 4 + \cos^2 \pi / 4$ = ----------

(A) 0
(B) -1
(C) 1
(D) $1/\sqrt{2}$

Q4. Sec (π/2 – θ) = ----------

(A) - sec θ
(B) - cosec θ
(C) sec θ
(D) cosec θ

Q5. Period of cosec x is ----------

(A) 2 π
(B) π
(C) 3 π
(D) π/2

Q6. Radius of escribed circle opposite to vertex A of triangle is ----------

(A) Δ/s (B) $\Delta/(s-a)$

(C) $\Delta/(s-b)$ (D) $\Delta/(s-c)$

Q7. Cos x = 1/2 , then x = ----------

(A) $\pi/6$ (B) $\pi/4$

(C) $\pi/3$ (D) $\pi/2$

Q8. Sin (Cos^{-1} $\sqrt{3}/2$) = ----------

(A) $\pi/6$ (B) $\pi/3$

(C) $2/\sqrt{3}$ (D) 1/2

Q9. 1 is not ---------- number

(A) Odd (B) Real

(C) Prime (D) Rational

Q10. Multiplicative inverse of complex number (0,1)

(A) (0 , -1) (B) (-1 , 0)

(C) (1 , 0) (D) (0 , 1)

Q11. Set G is closed and associative with respect to binary operation, then set G is called

(A) Groupoid (B) Semi-Group

(C) Monoid (D) Group

Q12. Disjunction of two statements p and q is

(A) $p \wedge q$ (B) $p \vee q$

(C) $p \rightarrow q$ (D) $p \leftrightarrow q$

Q13. Tabular form of $\{ x \mid x \in N \wedge x + 4 = 0 \}$ is

(A) { } (B) {0}

(C) {-4} (D) {0,4}

Q14. A square matrix A is symmetric if A^t =

(A) A^t (B) $- A^t$

(C) A (D) - A

Q15. If order of matrix A is 2 x 5 and order of B is 5 x 7 , then order of AB is ----------

(A) 5 x 2 (B) 7 x 5

(C) 7 x 2 (D) 2 x 7

Q16. α , β are roots of $x^2 + 2x + 1 = 0$, then $\alpha^2 + \beta^2$ = ----------

(A) 8 (B) 4

(C) -2 (D) 2

Q17. If ω is cube root of unity, then $(1 + \omega + \omega^2)^2$ = ----------

(A) ω (B) ω^2

(C) 0 (D) 1

Q18. $2/(x^2 - 1) = 1/(x - 1) + B/(x + 1)$, then value of B is

(A) 1 (B) -1

(C) 2 (D) -2

Q19. Sum the series 1+ 9/10 + 81/100 + ----------

(A) 10 (B) 9

(C) 9/10 (D) 10/9

Q20. 5th term of sequence whose general term is $a_n = n + (-1)^n$ is

(A) 4 (B) -4

(C) 5 (D) -5

SUBJECTIVE PART SECTION-I / SECTION-II

Time Allowed: 2 HRS 30 MINUTES | **Maximum Marks:** 80

Q2. (Marks 16) Write short answers any Eight (8) of the following

i Simplify (7, 9) + (3 , -5)

ii Find the multiplicative inverse of (-4,7)

iii ∀ z ∈ C , prove that $z.\bar{z} = |z|^2$

iv Simplify i^{-10}

v Write the power set of { 9 , 11 }

vi Construct the truth table for (p ∧ ~ p) → q

vii Find x and y if [x+3 1; -3 3y-4] = [2 1; -3 2]

viii If A and B are square matrices of the same order, then explain why in general $(A+B)^2 \neq A^2 + 2AB + B^2$

ix Without expansion show that the determinant of the matrix [6 7 8; 3 4 5; 2 3 4] = 0

x Solve the equation $x^2 - 2x - 899 = 0$ by completing the square

xi Evaluate $\omega^{28} + \omega^{29} + 1$

xii Find the condition that one root of equation $x^2 + px + q = 0$ is double the other.

Q3. (Marks 16) Write short answers any Eight (8) of the following

i Define an identity

ii Change $(6x^3+5x^2-7)/(2x^2-x-1)$ in to proper fraction

iii Find the next two terms 1 , 3 , 7 , 15 , 31 , --------

iv If $a_{n-3} = 2n-5$, find the nth term of the sequence

v Show that the reciprocals of the terms of the geometric sequence a_1 , a_1r^2, a_1r^4, ---------- form another geometric sequence

vi Find A.M between x - 3 and x + 5

vii Find the value of n when ${}^nP_4 : {}^{n-1}P_3 = 9 : 1$

viii Find the value of n when ${}^nC_{10} = (12 \times 11)/(2!)$

ix Determine the probability of getting 2 heads and 2 tails when a coin is tossed four times

x Prove $1 + 4 + 7 +$ --------- $+ (3n - 2) = n(3n-1)/2$

xi Calculate by means of Binomial theorem $(0.97)^3$

xii Expand $(8 - 5x)^{-2/3}$ up to four terms.

Q4. (Marks 18) Write short answers any Nine (9) of the following

i If $\tan\theta = 8/15$ and terminal arm of the angle is in the III quadrant, find the value of $\sin\theta$ and $\cos\theta$

ii Prove that $\sec^2\theta - \operatorname{cosec}^2\theta = \tan^2\theta - \cot^2\theta$

iii If α , β , γ are angles of a triangle ABC , Prove that $\tan(\alpha + \beta) + \tan\gamma = 0$

iv Find value of $\sec 75°$, without using tables

v Prove that $\cos 20^o + \cos 100^o + \cos 140^o = 0$

vi Write the domain and range of $y = \tan x$

vii Find the period of cosec 10x

viii Draw the graph of $y = \sin x/2$ for $0 \leq x \leq 2\pi$

ix Find the smallest angle of the triangle ABC , when a = 37.34 , b = 3.24 , c = 35.06

x Find area of triangle ABC , if a = 18 , b = 24 , c = 30

xi Prove that $r\, r_1\, r_2\, r_3 = \Delta^2$

xii Without using calculator , show that $2\cos^{-1} 4/5 = \sin^{-1} 24/25$

xiii Find the solution of equation $\operatorname{cosec}\theta = 2$ which lies in $[0 , 2\pi]$

Note: Note: Attempt any Three questions from this section 10 x 3 = 30

Q5.

(A) For what values of m , will the roots of the equation $x^2 - 2(1+3m)x + 7 (3 + 2m) = 0$ be equal

(B) Solve the system linear equations by Cramer's Rule: $2x_1 – x_2 + x_3 = 8$; $x_1 + 2x_2 + 2x_3 = 6$; $x_1 - 2x_2 - x_3 = 1$

Q6.

(A) Resolve into partial fractions $1/((1-ax)(1-bx)(1-cx))$

(B) If $y = 2/3\ x + 4/9\ x^2 + 8/27\ x^3 +$ ---------- and if $0 < x < 3/x$, then show that $x = 3y / (2(1+y))$

Q7.

(A) Prove that ${}^{n-1}C_r + {}^{n-1}C_{r-1} = {}^{n}C_r$

(B) If x is so small that its square and higher powers can be neglected, show that $(1-7x)/\sqrt{(1+x)} \approx 1 – 3/2\ x$

Q8.

(A) Show that $\cos 20^o \cos 40^o \cos 80^o = 1/8$

(B) By using $\Delta = 1/2\ bc \sin \alpha$ drive the Hero's formula

Q9.

(A) If $\cot\theta = 5/2$ and the terminal arm of the angle is in the I quad, find the value of $(3 \sin\theta + 4 \cos\theta)/(\cos\theta - \sin\theta)$

(B) Prove that $2 \tan^{-1} 1/3 + \tan^{-1} 1/7 = \pi/4$

GROUP – II

OBJECTIVE

Maximum Marks: 20 | **Paper Code:** 6198

Note: You have four choices for each objective type question as A , B , C and D . The choice which you think is correct, fill that circle in front of that question number. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question.

Q1. Number of necklaces can be made from 6 beads

(A) 720 (B) 120

(C) 90 (D) 60

Q2. Middle term in expansion of $(3 + x)^4$ is

(A) $81\ x^2$ (B) $54\ x^2$

(C) $26\ x^2$ (D) $108\ x^2$

Q3. One degree is equal to ..... radian

(A) 180/π (B) π/180

(C) π/90 (D) π

Q4. cot (90 - α) = ...

(A) tan α (B) - tan α

(C) cot α (D) - cot α

Q5. Period of sin x/3 is

(A) 2 π (B) 2π/3

(C) 6 π (D) 3 π

Q6. cos α/2 = ...

(A) s(s-a)/bc (B) s(s-b)/ac

(C) √(s(s-a)/bc) (D) √(s(s-b)/ac)

Q7. sec $(\cos^{-1} 1/2)$ = ----------

(A) 1/2 (B) 2

(C) π/3 (D) π/6

Q8. If cos x = - √3/2 , then value of x is

(A) 5π/6 (B) π/6

(C) π/3 (D) −π/3

Q9. a < b ⇒ - a > - b , a , b ∈ R property used is

(A) Transitive (B) Additive

(C) Multiplicative (D) Trichotomy

Q10. If Z = 1 – i , then |Z| = ...

(A) 2 (B) -2

(C) √−2 (D) √2

Q11. A and B are disjoint sets then

(A) $A \cap B = \emptyset$ (B) $A \cup B = \emptyset$

(C) $A - B = \emptyset$ (D) $B - A = \emptyset$

Q12. Tabular form of $\{ x \mid x \in E \wedge 2 < x \leq 4\}$

(A) {2 , 3 , 4} (B) {2 , 4}

(C) {4} (D) { ∅ }

Q13. The set A has m elements , Number of elements in power set of A

(A) 2^{m-1} (B) 2^m

(C) 2^{m+1} (D) $2^m/^2$

Q14. Rank of the matrix [1 1; 1 1] is

(A) Zero (B) 1

(C) - 1 (D) 2

Q15. Determinant of [-5] is

(A) Zero (B) Not possible

(C) - 5 (D) 5

Q16. α , β are roots of $ax^2 – bx + c = 0$, then α + β = ----------

(A) b/a (B) - b/a

(C) c/a (D) - c/a

Q17. If polynomial $x^2 – 2x + 2$ is divided by x – 1 , then remainder is

(A) -1 (B) 1

(C) 0 (D) 2

Q18. Partial fraction of $x/((x-1)(x+2)) = 1/(3(x-1)) + B/(x+2)$, then value of B is

(A) -3/2 (B) 3/2

(C) 2/3 (D) -2/3

Q19. Sum of n-arithmatic means between a and b is

(A) (a+b)/2 (B) n(a+b)

(C) (a+b) (D) n((a+b)/2)

Q20. Next term of sequence 7 , 9 , 12 , ... Is

(A) 14 (B) 15

(C) 16 (D) 18

SUBJECTIVE PART SECTION-I / SECTION-II

Time Allowed: 2 HRS 30 MINUTES | **Maximum Marks:** 80

Q2. (Marks 16) Write short answers any Eight (8) of the following

i Simplify (5, -4) (-3 , -2)

ii Separate into real and imaginary parts (2-7i)/(4+5i)

iii Prove that $\bar{Z} = Z$ if Z is real

iv Simplify $(a+b\,i)^2$

v Write two proper subsets of { a , b , c }

vi Show that $(p \wedge q) \rightarrow p$ is a tautology

vii Find x and y if [2 0 x; 1 y 3] + 2[1 x y; 0 2 -1] = [4 -2 3; 1 6 1]

viii Find the matrix X if [5 2; -2 1] X = [2 1; 5 10]

ix If A = [1 2 -3; 0 -2 0; -2 -2 1] , then find A_{12} and A_{32}

x Evaluate $\omega^{28} + \omega^{29} + 1$

xi Use remainder theorem to find the remainder when $x^2 + 3x + 7$ is divided by x + 1

xii Discuss the nature of the roots of equation $2x^2 - 5x + 1 = 0$

Q3. (Marks 16) Write short answers any Eight (8) of the following

i Define partial fraction resolution

ii Suppose (7x+25)/((x+3)(x+4)) = A/(x+3) + B/(x+4) Find the values of A and B

iii Write the first four terms of the following sequence , if $a_n = (-1)^n n^2$

iv Which term of the A.P 5 , 2 , - 1 , ... is - 85 ?

v If 1/a , 1/b and 1/c are in G.P. Show that the common ratio is $\pm \sqrt{(a/c)}$

vi Show that $G^2 = AH$ if a = 2 i , b = 4 i

vii Find the value of n if ${}^nP_2 = 30$

viii Find the number of the diagonals of a 6-sided figure

ix A die is rolled. What is the probability that the dots on the top are greater than 4 ?

x Prove that $4^k > 3^k + 4$ is true for k = 2 , 3

xi Calculate $(0.97)^3$ by means of binomial theorem

xii Expand up to 4 terms $(1 - x)^{1/2}$, taking the values of x such that the expansion is valid

Q4. (Marks 18) Write short answers any Nine (9) of the following

i Find ℓ , when θ = 65° 20′ , r = 18 mm

ii Verify that 2 sin 45° + 1/2 cosec 45° = 3/√2

iii Without using the tables , find the value of sec (- 300)

iv Prove that (cos 8° – sin 8°)/(cos 8° + sin 8°) = tan 37°

v Prove that 1 + tan α tan 2 α = sec 2 α

vi Write down the domain and range of sin x

vii Find the period of cot x/2

viii Draw the graph of y = cos x for 0 ≤ x ≤ 360°

ix What is difference between right angle triangle and oblique triangle

x Find the area of the triangle ABC , if a = 200 , b = 120 , γ = 150°

xi Find the radius of in-circle if a = 13 , b = 14 , c = 15

xii Without using calculator , show that $\tan^{-1} 5/12 = \sin^{-1} 5/13$

xiii Solve the equation sin x + cos x = 0

> Note: Note: Attempt any Three questions from this section 10 x 3 = 30

Q5.

(A) Solve the equation $\sqrt{(5x^2 + 7x + 2)} - \sqrt{(4x^2 + 7x + 18)} = x - 4$

(B) Use matrices to solve the following system of equation: $2x_1 + x_2 + 3x_3 = 3$; $x_1 + x_2 - 2x_3 = 0$; $-3x_1 - x_2 + x_3 = -4$

Q6.

(A) Resolve the following into partial fractions $x^2/((x-2)(x-1)^2)$

(B) Find n so that $(a^n + b^n)/(a^{n-1} + b^{n-1})$ may be the A.M. between a and b

Q7.

(A) A natural number is chosen out of the first fifty natural numbers. What is the probability that the chosen number is multiple of 3 or 5 ?

(B) Expand $(x/2 - 2/x^2)^6$ by using binomial theorem

Q8.

(A) Show that cos 20° cos 40° cos 80° = 1/8

(B) The sides of triangle are $x^2 + x + 1$, 2x + 1 and $x^2 - 1$. Prove that the greatest angle of the triangle is 120°

Q9.

(A) Prove that: $\sqrt{[(1-\sin\theta)/(1+\sin\theta)]} = \sec\theta - \tan\theta$, where θ is not an odd multiple of $\pi/2$

(B) Prove that: $\cos^{-1} A + \cos^{-1} B = \cos^{-1} [AB - \sqrt{(1 - A^2)}\sqrt{(1 - B^2)}]$